



Power Electronics (EE3032)

Sessional-II Exam

Date: April 09, 2025
Course Instructor(s)
Dr. Muhammad Naeem

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 3



Roll No

Section

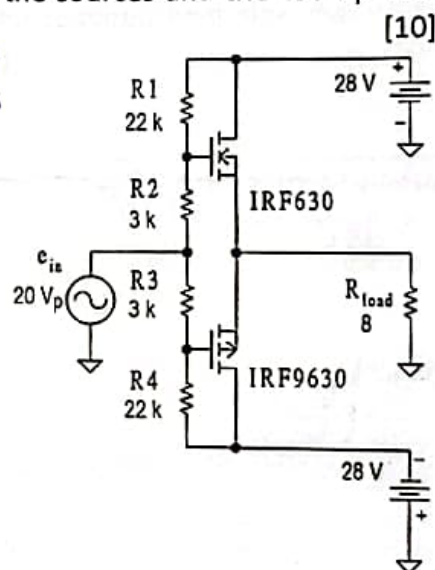
Student Signature

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Attempt all the questions.

CLO 2: Analyze audio power amplifier circuits.

Q1: Analyze the following audio amplifier circuit and calculate rms voltage at the gate of each switch, rms load voltage, power delivered to the load, power supplied by the sources and the total power dissipation in the amplifier.



CLO 3: Design power switches.

Q2: Design a switch using an IRF630 to drive a 25Ω load powered from a 150 V dc supply and determine the drain current, required gate pulse amplitude, power delivered to the load, the transistor power dissipation for an 80% duty cycle pulse, and the required heat sink maximum thermal resistance for $T_A = 60^\circ\text{C}$. ($R_{on @ 60^\circ\text{C}} = 0.4 \Omega$, $\theta_{JC} = 3 \frac{^\circ\text{C}}{\text{W}}$, $\theta_{CS} = 0.5 \frac{^\circ\text{C}}{\text{W}}$)

CLO 3: Design power switches.

Q3: Design a switch using a 2N3904 to drive a 20Ω load powered from a 10 V dc supply and determine the collector current, base current, base resistor, power delivered to the load, and the transistor power dissipation for a 70% duty cycle pulse.